Preetha Chatterjee

Assistant Professor · Drexel University

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Research Interests

Software engineering, software security, AI for software engineering, mining software repositories, software analytics, empirical software engineering, software evolution and maintenance, automated program repair, human and social aspects of software engineering

Professional Experience

2021-Present	**Assistant Professor**, *Department of Computer Science*, Drexel University, USA
2016-2021	**Research Assistant**, *Department of Computer and Information Sciences*, University of Delaware, USA
2014-2016	**Teaching Assistant**, *Department of Computer and Information Sciences*, University of Delaware, USA
2013-2014	**QA Engineer**, Integral Development Corporation, India
2012-2013	**Software Engineer**, Theorem Private Limited, India
2009-2012	**Project Engineer**, Wipro Technologies, India

Education

University of Delaware, DE, USA *2021*
Ph.D. in Computer Science
Department of Computer & Information Sciences
Dissertation Title: *Mining Information from Developer Chats Towards Building Software Maintenance Tools*
Advisor: Dr. Lori Pollock

University of Delaware, DE, USA *2016*
M.S. in Computer Science
Department of Computer & Information Sciences

West Bengal University of Technology, India *2008*
B.Tech. in Information Technology
Department of Information Technology

Honors and Awards

2026. **Faculty Research Excellence Award**, *Drexel College of Computing and Informatics*, in recognition of excellence, achievement, leadership, and innovation in research.

2024. **Outstanding Service to the Conference Organization Committee**, *21st International Conference on Mining Software Repositories (MSR 2024)*, in recognition of co-chairing the Mining Challenge Track of the conference.

2024. **CRA-WP Scholarship**, *Computing Research Association*, to attend Career Mentoring Workshop for early-career faculty.

2023. **ACM SIGSOFT Distinguished Reviewer Award**, *ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) 2023*, in recognition of delivering outstanding peer reviews to research track paper submissions.

2022. **Summer Research Award for Tenure/Tenure-track Faculty**, *Drexel University*, funding for most meritorious research proposals to pursue research that will enhance recipients' careers and increase their contributions to Drexel.

2021. **Frank A. Pehrson Graduate Student Award for Outstanding Computer Science Research**, *University of Delaware*, in recognition of outstanding performance and future potential in the field.

2020. **GHC Sponsorship**, *University of Delaware*, to attend virtual Grace Hopper Celebration of Women in Computing.

2019. **NSF Student Travel Support**, *U.S. National Science Foundation*, to financially support selected researchers in attending the 41st International Conference on Software Engineering (ICSE).

2017. **ACM-SIGSOFT CAPS Travel Award**, *ACM Special Interest Group on Software Engineering*, to financially support selected researchers in attending the 14th International Conference on Mining Software Repositories (MSR).

2017. **Professional Development Award**, *University of Delaware*, to help selected graduate students participate in significant professional conferences pertaining to their field of study.

2017. **WIE Conference Travel Scholarship**, *Women in Engineering Association at University of Delaware*, to help selected graduate students participate in significant professional conferences.

2017. **CRA-W Travel Scholarship**, *Computing Research Association*, to attend Grad Cohort Workshop.

2016. **Lauri Pfeffer Shinn Memorial Award**, *University of Delaware*, in recognition of academic success and contribution to the Department of Computer and Information Sciences.

2015. **GHC Travel Scholarship**, *University of Delaware*, to attend Grace Hopper Celebration of Women in Computing as a mentor to undergraduate students.

2015. **CRA-W Travel Scholarship**, *Computing Research Association*, to attend Grad Cohort Workshop.

Research Funding

NSF CyberAI SFS Program

Title: "CyberAI Innovation: Preparing AI-Native Software Engineers for Secure Software Development"
Lead PI: **Preetha Chatterjee** at Drexel University
Non-lead PI: Kostadin Damevski at Virginia Commonwealth University
Award Number: 2622910 (Start Date: 09/01/2026 - End Date: 08/31/2029)
Total Award Amount: $499,979, **Drexel's share (my share)**: $249,979

NSF Finders Foundry Program

Title: "NSF FF: Planning: Co-Designing an AI-Supported Empathic Communication Platform for High School Classrooms"
PI: **Preetha Chatterjee**, School of Computer and Information Sciences, Drexel University
co-PI: Jennifer Katz-Buonincontro, School of Education, Drexel University
Award Number: 2627511 (Start Date: 07/15/2026 - End Date: 10/31/2026)
Total Award Amount: $49,861, **Drexel SCIS' share (my share)**: $36,014

OpenAI CyberSecurity Grant Program

Title: "Proactive Software Security with Agentic AI"
Sole PI: **Preetha Chatterjee**
Award Start Date: 08/12/2025 - End Date: 06/30/2026
Total Award Amount: $3,000 in API credits

OpenAI Researcher Access Program

Title: "LLM-guided bug resolution"
Sole PI: **Preetha Chatterjee**
Award Start Date: 06/30/2025 - End Date: 05/30/2026
Total Award Amount: $1,000 in API credits

Drexel University Faculty Summer Research Awards for Tenure/Tenure-Track Faculty

Title: "Towards Enabling Affective Awareness in Free Open Source Software (FOSS)"
Sole PI: **Preetha Chatterjee**
Award Start Date: 07/01/2022 - End Date: 06/30/2023
Total Award Amount: $7,000, **Drexel SCIS' share (my share)**: $7,000

Publications

[Underlined Students are advised by me at Drexel University]

Refereed Journal Publications

[J9] Ramtin Ehsani, Shriya Rawal, Yuanfang Cai, and **Preetha Chatterjee**, "Faster Code, Deeper Debt? A Multivocal Literature Review on Technical Debt and Its Early Signs in LLM-Assisted Software Development", Transactions on Software Engineering and Methodology (TOSEM) (Impact factor: 6.2) .

[J8] Emily R. Woodland, Giselle C. Matlis, **Preetha Chatterjee**, Thomas C. Palazzolo, Vakhtang Tchantchaleishvili, Marvin J. Slepian, and Amy L. Throckmorton, "Artificial Intelligence in the Advancement of Mechanical Circulatory Support and Cardiac Care: A Systematic Review of its Role and Potential", Artificial Organs, 2026 (Impact factor: 2.3).

[J7] Evan M. Forman, Charlotte J. Hagerman, Asher E. Hong, Zhuoran Huang, Hannah C. McCausland, Jasmine H. Sun, Lauren C. Taylor, Meghan L. Butryn, **Preetha Chatterjee**, Ramtin Ehsani, Nawal Syed, "Feasibility Study of an AI Chatbot-Powered Cognitive-Behavioral Lifestyle Modification Program", Annals of Behavioral Medicine, Volume 60, Issue 1, 2026 (Impact factor: 4.9) .

[J6] Amirali Sajadi, Kostadin Damevski, and **Preetha Chatterjee**, "Psycholinguistic Analyses in Software Engineering Text: A Systematic Literature Review", Information and Software Technology (IST), Volume 189, January 2026 (Impact factor: 4.3).

[J5] Ramtin Ehsani, Sakshi Pathak, Esteban Parra Rodriguez, Sonia Haiduc, and **Preetha Chatterjee**, "What Characteristics Make ChatGPT Effective for Software Issue Resolution? An Empirical Study of Task, Project, and Conversational Signals in GitHub Issues", Empirical Software Engineering (EMSE), Volume 31, article number 22 (2026) (Impact factor: 3.6). *Presented as a Journal First Paper at ASE '2026*.

[J4] Amirali Sajadi, Binh Le, Anh Nguyen, Kostadin Damevski, and **Preetha Chatterjee**, "Do LLMs Consider Security? An Empirical Study on Responses to Programming Questions", Empirical Software Engineering (EMSE), Volume 30, article number 101, (2025) (Impact factor: 3.6)

[J3] Layla Bouzoubaa, Ramtin Ehsani, **Preetha Chatterjee**, and Rezvaneh (Shadi) Rezapour, "Shifting Narratives in Media Coverage: A Decade of Drug Discourse in The Philadelphia Inquirer", Journal of Medical Internet Research (JMIR) Infodemiology, Vol 5 (2025) (Impact factor: 3.5).

[J2] **Preetha Chatterjee**, Kostadin Damevski, Nicholas A. Kraft, and Lori Pollock, "Automatically Identifying the Quality of Developer Chats for Post Hoc Use", Transactions on Software Engineering and Methodology (TOSEM), Volume 30, Issue 4, October 2021 (Impact factor: 4.4). *Presented as a Journal First Paper at ASE '2022*.

[J1] **Preetha Chatterjee**, Minji Kong, and Lori Pollock, "Finding Help with Programming Errors: An Exploratory Study of Novice Software Engineers' Focus in Stack Overflow Posts", Journal of Systems and Software (JSS), Volume 159, January 2020 (Impact factor: 3.7).

Refereed Conference Publications

[C25] Ramtin Ehsani, Irene Manotas, Saurabh Pujar, Luca Buratti, and **Preetha Chatterjee**, "How Do LLMs Read GitHub Bug Reports? An Attention Analysis of LLMs for Automated Program Repair", in Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE), Research Papers Track, Nov 2026. (Acceptance rate: 20%)

[C24] Mia Mohammad Imran, Robert Zita, Rahat Rizvi Rahman, **Preetha Chatterjee**, and Kostadin Damevski, "Toxicity Ahead: Forecasting Conversational Derailment on GitHub", in Proceedings of the 48th International Conference on Software Engineering (ICSE), Research Track, Apr 2026. (Acceptance rate: 19.8%).

[C23] Ramtin Ehsani, Sakshi Pathak, Shriya Rawal, Abdullah Al Mujahid, Mia Mohammad Imran, and **Preetha Chatterjee**, "Where Do AI Coding Agents Fail? An Empirical Study of Failed Agentic Pull Requests in GitHub", in Proceedings of the 23rd International Conference on Mining Software Repositories (MSR), Mining Challenge Track, Apr 2026.

[C22] Amirali Sajadi, Tu Nguyen, Kimmie Huynh, Esteban Parra, and **Preetha Chatterjee**, "TraceView: Interactive Visualization of Agentic Program Repair Trajectories", in Proceedings of the 42nd International Conference on Software Maintenance and Evolution (ICSME), Tool Demonstration and Data Showcase Track, Sept 2026.

[C21] Esteban Parra, Sonia Haiduc, **Preetha Chatterjee**, Ramtin Ehsani, and Polina Iaremchuk, "Towards A Sustainable Future for Peer Review in Software Engineering", in Proceedings of the 48th International Conference on Software Engineering (ICSE), Future of Software Engineering (FoSE) Track, Apr 2026.

[C20] Ramtin Ehsani, Esteban Parra Rodriguez, Sonia Haiduc, and **Preetha Chatterjee**, "Hierarchical Knowledge Injection for Improving LLM-based Program Repair", in Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE), Research Papers Track, Nov 2025 (Acceptance rate without revisions: 9.4%).

[C19] Ramtin Ehsani, Sakshi Pathak, and **Preetha Chatterjee**, "Towards Detecting Prompt Knowledge Gaps for Improved LLM-guided Issue Resolution", in Proceedings of the 22nd International Conference on Mining Software Repositories (MSR), Technical Track, Apr 2025 (Acceptance rate: 27%).

[C18] Yousab Grees, Polina Iaremchuk, Ramtin Ehsani, Esteban Parra, **Preetha Chatterjee**, and Sonia Haiduc, "AI-Powered Commit Explorer (APCE)", in Proceedings of the 41st International Conference on Software Maintenance and Evolution (ICSME), Tool Demonstration Track, Sept 2025.

[C17] Ramtin Ehsani, Mia Mohammad Imran, Robert Zita, Kostadin Damevski, and **Preetha Chatterjee**, "Incivility in Open Source Projects: A Comprehensive Annotated Dataset of Locked GitHub Issue Threads", in Proceedings of the 21st International Conference on Mining Software Repositories (MSR), Data Showcase Track, Apr 2024.

[C16] Shyamal Mishra and **Preetha Chatterjee**, "Exploring ChatGPT for Toxicity Detection in GitHub", in Proceedings of the 46th International Conference on Software Engineering (ICSE), New Ideas and Emerging Results Track, Apr 2024 (Acceptance rate: 22%).

[C15] Mia Mohammad Imran, **Preetha Chatterjee**, and Kostadin Damevski, "Shedding Light on Software Engineering-specific Metaphors and Idioms", in Proceedings of the 46th International Conference on Software Engineering (ICSE), Research Track, Apr 2024 (Acceptance rate: 18%).

[C14] Mia Mohammad Imran, **Preetha Chatterjee**, and Kostadin Damevski, "Uncovering the Causes of Emotions in Software Developer Communication Using Zero-shot LLMs", in Proceedings of the 46th International Conference on Software Engineering (ICSE), Research Track, Apr 2024 (Acceptance rate: 18%).

[C13] Amirali Sajadi, Kostadin Damevski, and **Preetha Chatterjee**, "Interpersonal Trust in OSS: Exploring Dimensions of Trust in GitHub Pull Requests", in Proceedings of the 45th International Conference on Software Engineering (ICSE), New Ideas and Emerging Results Track, May 2023 (Acceptance rate: 22%).

[C12] Ramtin Ehsani, Rezvaneh Rezapour, and **Preetha Chatterjee**, "Exploring Moral Principles Exhibited in OSS: A Case Study on GitHub Heated Issues", in Proceedings of the 31st ACM Joint European Software Engineering Conference

and Symposium on the Foundations of Software Engineering (ESEC/FSE), Ideas, Visions and Reflections Track, Dec 2023.

[C11] Amirali Sajadi, Kostadin Damevski, and **Preetha Chatterjee**, "Towards Understanding Emotions in Informal Developer Interactions: A Gitter Chat Study", in Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Ideas, Visions and Reflections Track, Dec 2023.

[C10] Layla Bouzoubaa, Ramtin Ehsani, **Preetha Chatterjee**, and Rezvaneh Rezapour, "The Evolution of Substance Use Coverage in the Philadelphia Inquirer", in Proceedings of the 17th International AAAI Conference On Web And Social Media (ICWSM), Data Challenge, Jun 2023.

[C9] Mia Mohammad Imran, Yashasvi Jain, **Preetha Chatterjee**, and Kostadin Damevski, "Data Augmentation for Improving Emotion Recognition in Software Engineering Communication", in Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (ASE), Research Track, Oct 2022 (Acceptance rate: 18%).

[C8] Keerthana Muthu Subash, Lakshmi Prasanna Kumar, Sri Lakshmi Vadlamani, **Preetha Chatterjee**, and Olga Baysal, "DISCO: A Dataset of Discord Chat Conversations for Software Engineering Research", in Proceedings of the 19th International Conference on Mining Software Repositories (MSR), Data Showcase Track, May 2022.

[C7] **Preetha Chatterjee**, Tushar Sharma, and Paul Ralph, "Empirical Standards for Repository Mining", in Proceedings of the 19th International Conference on Mining Software Repositories (MSR), Tutorial, May 2022

[C6] **Preetha Chatterjee**, Kostadin Damevski, and Lori Pollock, "Automatic Extraction of Opinion-based Q&A from Online Developer Chats", in Proceedings of the 43rd International Conference on Software Engineering (ICSE), Technical Track, May 2021 (Acceptance Rate: 22.4%).

[C5] **Preetha Chatterjee**, Kostadin Damevski, Nicholas A. Kraft, and Lori Pollock, "Software-related Slack Chats with Disentangled Conversations", in Proceedings of the 17th International Conference on Mining Software Repositories (MSR), Data Showcase Track, May 2020 (Acceptance Rate: 38.8%).

[C4] **Preetha Chatterjee**, "Extracting Archival-Quality Information from Software-Related Chats", in Proceedings of the 42nd International Conference on Software Engineering (ICSE), Doctoral Symposium Track, May 2020 (Invited Talk Acceptance rate: 34.6%).

[C3] **Preetha Chatterjee**, Kostadin Damevski, Lori Pollock, Vinay Augustine, and Nicholas A. Kraft, "Exploratory Study of Slack Q&A Chats as a Mining Source for Software Engineering Tools", in Proceedings of the 16th International Conference on Mining Software Repositories (MSR), Research Track, May 2019 (Acceptance rate: 25.4%).

[C2] **Preetha Chatterjee**, Benjamin Gause, Hunter Hedinger, and Lori Pollock, "Extracting Code Segments and Their Descriptions from Research Articles," in Proceedings of the 14th International Conference on Mining Software Repositories (MSR), Research Track, May 2017 (Acceptance rate: 30.6%).

[C1] **Preetha Chatterjee**, Manziba Akanda Nishi, Kostadin Damevski, Vinay Augustine, Lori Pollock, and Nicholas A. Kraft, "What Information about Code Snippets Is Available in Different Software-Related Documents? An Exploratory Study", in Proceedings of the 24th IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER), Early Research Achievements Track, Feb 2017 (Acceptance rate: 32.5%).

Refereed Workshop Publications

[W2] Ramtin Ehsani, Rezvaneh (Shadi) Rezapour, **Preetha Chatterjee**, "Analyzing Toxicity in Open Source Software Communications Using Psycholinguistics and Moral Foundations Theory", in Proceedings of the 4th International Workshop on Natural Language-based Software Engineering (NLBSE), co-located with ICSE, Apr 2025.

[W1] **Preetha Chatterjee**, "Automatic Identification of Informative Code in Stack Overflow Posts", in Proceedings of the 1st International Workshop on Natural Language-based Software Engineering (NLBSE), co-located with ICSE, May 2022.

Extended abstracts and Symposiums

[E2] Layla Bouzoubaa, Ramtin Ehsani, **Preetha Chatterjee**, and Rezvaneh (Shadi) Rezapour, "Shifting Narratives in Media Coverage: A Decade of Drug Discourse in The Philadelphia Inquirer", 10th International Conference on Computational Social Science (IC2S2).

[E1] Evan M. Forman, Meghan L. Butryn, **Preetha Chatterjee**, Ramtin Ehsani, Charlotte J. Hagerman, Asher E. Hong, Zhuoran Huang, Hannah C. McCausland, Jasmine Sun, Lauren C. Taylor, "Feasibility Study of an AI Chatbot-Powered Cognitive-Behavioral Lifestyle Modification Program", Annual Meeting of the Eating Disorders Research Society (EDRS).

Other Manuscripts

[M2] **Preetha Chatterjee**, "Mining Information from Developer Chats Towards Building Software Maintenance Tools", University of Delaware, Ph.D. Thesis, 2021.

[M1] **Preetha Chatterjee**, "Exploring the Generality of a Java-based Loop Action Model for the Quorum Programming Language", University of Delaware, Ph.D. Prelims, 2015.

Talks and Presentations

Invited Talks

2026. *Post-PhD Panel: What is my life like as a computing researcher?*,
Invited Panelist, UR2PhD Undergraduate Mentoring Workshop, Computing Research Association (CRA), USA

2025. *Incivility in Open Source Projects*,
Invited Lecture, Colorado State University, USA

2023. *Emotion Awareness in Software Engineering*,
Lightning Talk, It Will Never Work in Theory (NWiT) April, 2023 series, Virtual.

2022. *Mining Software Repositories*,
Comcast Software Engineering Research Presentation, Drexel University, USA.

2022. *ACM SIGSOFT Empirical Standards of Mining Software Repositories*,
Tutorial at the 19th International Conference on Mining Software Repositories (MSR).

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Keynote Talk at Drexel Society of Artificial Intelligence, Drexel University, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at University of Maryland Baltimore County, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at Carleton University, Canada.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at Auburn University, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at Drexel University, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at Oakland University, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*,
Seminar at Boise State University, USA.

2021. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*, Seminar at George Mason University, USA.

2020. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*, Seminar at Inria, France.

2020. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*, Seminar at University of Zurich, Switzerland.

2020. *Mining Software Developer Chats Towards Improving Software Maintenance Tools*, Seminar at Delft University of Technology (TU Delft), Netherlands.

2017. *Mining Code Examples with Descriptive Text from Software Artifacts*, Seminar at the Special Interest Group in Software Systems at University of Delaware, USA.

Research Presentations

2026. *How Do LLMs Read Bug Reports? An Empirical Study of Attention in LLMs for Automated Program Repair*, 41st IEEE/ACM International Conference on Automated Software Engineering (ASE), Virtual Research Paper Presentation Talk, Munich, Germany.

2025. *Hierarchical Knowledge Injection for Improving LLM-based Program Repair*, 40th IEEE/ACM International Conference on Automated Software Engineering (ASE), Virtual Research Paper Presentation Talk, Seoul, South Korea.

2024. *Exploring ChatGPT for Toxicity Detection in GitHub*, 46th International Conference on Software Engineering (ICSE), Paper Presentation Talk, Lisbon, Portugal.

2022. *Automatically Identifying the Quality of Developer Chats for Post Hoc Use*, 37th IEEE/ACM International Conference on Automated Software Engineering (ASE), Journal-first Paper Presentation, Virtual.

2022. *Automatic Identification of Informative Code in Stack Overflow Posts*, International Workshop on Natural Language-based Software Engineering (NLBSE), Paper Presentation (co-located ICSE events), Virtual.

2021. *Automatic Extraction of Opinion-based Q&A from Online Developer Chats*, Journal of Systems and Software (JSS), Invited Paper Video, Virtual.

2021. *Mining Information from Developer Chats Towards Building Software Maintenance Tools*, University of Delaware, PhD Thesis Dissertation, Newark, USA.

2021. *Automatic Extraction of Opinion-based Q&A from Online Developer Chats*, 43rd International Conference on Software Engineering (ICSE), Paper Presentation Talk, Virtual.

2021. *Finding Help with Programming Errors: An Exploratory Study of Novice Software Engineers' Focus in Stack Overflow Posts*, Journal of Systems and Software (JSS), Featured Paper Video, Virtual.

2020. *Extracting Archival-Quality Information from Software-Related Chats*, 42nd International Conference on Software Engineering (ICSE), Paper Presentation Talk, Virtual.

2020. *Software-related Slack Chats with Disentangled Conversations*, 17th International Conference on Mining Software Repositories (MSR), Paper Presentation, Virtual.

2019. *Exploratory Study of Slack Q&A Chats as a Mining Source for Software Engineering Tools*, 16th International Conference on Mining Software Repositories (MSR), Paper Presentation Talk, Montreal, Canada.

2019. *Extracting Archival-Quality Information from Software-Related Chats*, University of Delaware, PhD Proposal Presentation Talk, Newark, USA.

2018. *Finding My Solution in the Haystack of a Post: An Exploratory Study of Stack Overflow*, CIS Department 55 Years Anniversary Showcase at University of Delaware, Poster Presentation, Newark, USA.

2017. *Extracting Code Segments and Their Descriptions from Research Articles*, 14th International Conference on Mining Software Repositories (MSR), Paper Presentation Talk, Buenos Aires, Argentina.

2017. *What Information about Code Snippets Is Available in Different Software-Related Documents? An Exploratory Study*, CRA-W Grad Cohort Workshop, Poster Presentation, Washington D.C., USA.

2015. *Exploring the Generality of a Java-based Loop Action Model for the Quorum Programming Language*, University of Delaware, PhD Prelims Presentation Talk, Newark, USA.

Teaching Experience

Department of Computer Science, Drexel University

Fall 2026	**SE 201: Introduction to Software Engineering and Development**, Instructor	*Enrollment:46*
Spring 2026	**SE 611: Software Analytics (2 sections)**, Instructor	*Enrollment:15*
Fall 2025	**SE 201: Introduction to Software Engineering and Development (2 sections)**, Instructor	*Enrollment:63*
Fall 2024	**SE 201: Introduction to Software Engineering and Development**, Instructor	*Enrollment:39*
Fall 2023	**SE 181: Introduction to Software Engineering and Development (2 sections)**, Instructor	*Enrollment:115*
Spring 2023	**CS T680: Software Analytics (2 sections)**, Instructor	*Enrollment:13*
Fall 2022	**SE 181: Introduction to Software Engineering and Development**, Instructor	*Enrollment:55*
Spring 2022	**SE 181: Introduction to Software Engineering and Development**, Instructor	*Enrollment:55*
Fall 2021	**SE 181: Introduction to Software Engineering and Development**, Instructor	*Enrollment:85*

Department of Computer and Information Sciences, University of Delaware

Summer 2019	**CISC 181: Introduction to Computer Science II**, Instructor
Fall 2018	**CISC 367: Intro to Computer Science Research**, Substitute Instructor
Spring 2018	**CISC 667: Communication Skills for CS Researchers**, Substitute Instructor
Fall 2017	**CISC 879: Text Analysis for Software Engineering**, Substitute Instructor
Spring 2016	**CISC 474: Advanced Web Technologies**, Teaching Assistant
Fall 2015	**CISC 103: Web Applications using Computer Science**, Teaching Assistant
Spring 2015	**CISC 106: General Computer Science for Engineers**, Teaching Assistant
Fall 2014	**CISC 181: Introduction to Computer Science II**, Teaching Assistant

Student Research Mentoring

Dissertation Committee Chair

WI'23-Present	**Ramtin Ehsani**, Ph.D. in CS, Drexel University (Passed Ph.D. Candidacy in 2024)
FA'22-Present	**Amirali Sajadi**, Ph.D. in CS, Drexel University (Passed Ph.D. Candidacy in 2024)

Dissertation/Proposal/Candidacy Committee Member

2026 **Baojuan Wang**, Drexel University, `Candidacy Committee`

2024 **Ernie Pisch**, Drexel University, `Candidacy Committee`

2023-2024 **Mia Mohammed Imran**, VCU, `Proposal and Dissertation Committee`

2023 **Chaewon Yun**, Drexel University, `Candidacy Committee`

2022-2024 **Sergey Matskevich**, Drexel University, `Proposal and Dissertation Committee`

2022-2026 **Jason Lefever**, Drexel University, `Candidacy, Proposal, Dissertation Committee`

Master's Students

FA'26-Present **Priti Sagar**, Drexel University (Independent Study)
Secure Agentic Software Development

SP'26-SU'26 **Mokshad Sankhe**, Drexel University
Mining the Evolution of AI-generated code

WI'25-Present **Shriya Rawal**, Drexel University
Understanding Quality of AI-generated code– *[Co-author of papers at MSR 2026, TOSEM 2026]*

WI'24-SP'25 **Sakshi Pathak**, Drexel University (Co-op Program)
Analysis of software developer-LLM collaboration – *[Co-author of papers at MSR 2025, MSR 2026, EMSE 2026]*

FA'23 **Allison Gong**, Drexel University
Exploring role of LLMs in SE Education

SU'23 **Shyamal Mishra**, Drexel University (Co-op program)
Chatbot-based applications in software engineering – *[Co-author of paper at ICSE 2024]*

Undergraduate Students

FA'26-present **Moazzam Mazen**, Drexel University
AI for SE

FA'25-present **John Nguyen**, Drexel University
Securing agentic AI software systems *[Co-author of paper at ICSME 2026]*

WI'26-SU'26 **Amoz Lee**, Drexel University
Understanding human-AI collaboration for SE

WI'26-SP'26 **Kimmie Huynh**, Drexel University
Securing agentic AI software systems *[Co-author of paper at ICSME 2026]*

SP'25-SU'25 **Moon Mustafa**, Drexel University
Enhancing LLM-assisted bug fixing and issue resolution

SU'24 **Taqi Tahmid**, Drexel University
Understanding security concerns in LLM responses to programming questions

SP'24-SU'24 **Binh Le**, Drexel University
Security concerns in LLM code responses – *[Co-author of paper at EMSE 2025]*

SP'24-SU'24 **Anh Nguyen**, Drexel University
Security concerns in LLM code responses – *[Co-author of paper at EMSE 2025]*

SP' 2024 **Giles Odigwe**, Drexel University (Co-op Program)
Analysis of LLM responses to programming questions

WI' 2024 **Mustafa Bookwala**, Drexel University
Analysis of software developer-ChatGPT conversations

WI' 2023 **Vanessa Martinez**, Drexel University (Pathway to SuperNova Research Fellows program)
Analysis of human and social aspects in software engineering

WI' 2022 **Thomas Do**, Drexel University (Pathway to SuperNova Research Fellows program)
Analysis of human aspects in software engineering

Yashasvi Jain, Drexel University
2021-2022 Analysis of emotions in developer textual communications – *[Co-author of paper at ASE 2022]*

2020 **Kevin Mason**, University of Delaware
Data analysis of software developer chat conversations

2019-2020 **Brian Phillips**, University of Delaware
Data analysis and case study of software developer communications

2019-2020 **Humpher Owusu**, University of Delaware
Data analysis and case study of software developer communications

Minji Kong, University of Delaware
2018 Qualitative analysis of developers' focus in Stack Overflow – *[Co-author of paper at JSS 2020]*

2017 **Qilin Ma**, University of Delaware
Research tool development for mining software-related Q&A forums

2016 **Benjamin Gause**, University of Delaware
Automatic extraction of code descriptions – *[Co-author of paper at MSR 2017]*

2016 **Hunter Hedinger**, University of Delaware
Automatic extraction of code descriptions – *[Co-author of paper at MSR 2017]*

Professional Service

Steering Committee

2026-Present Steering Committee Member, Intl. Conf. on Mining Software Repositories (MSR)

Journal Editorial Board

2021 - 2024 Publicity Chair, Journal of Systems and Software (JSS)

Organizing Committee

2027 *Program Committee* Co-Chair, 24th Intl. Conf. on Mining Software Repositories (MSR)

2025 *Tutorials Track* Co-Chair, 22nd Intl. Conf. on Mining Software Repositories (MSR)

2024 *Journal-first* Co-Chair, 32nd IEEE/ACM Intl. Conf. on Program Comprehension (ICPC)

2024 *Mining Challenge* Co-Chair, 21st Intl. Conf. on Mining Software Repositories (MSR)
[Outstanding Service Award] 🏆

2023 *NIER* PC co-Chair, 23rd IEEE Intl. Conf. on Source Code Analysis and Manipulation (SCAM)

2023 PC co-Chair, 3rd Intl. Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems/Internet of Things (SEA4DQ) *at FSE'23*

2023 Diversity and Inclusion co-Chair, Intl. Conf. on Mining Software Repositories (MSR)

2022 PC co-Chair, 1st Intl. Workshop on Recruiting Participants for Empirical SE (RoPES) *at ICSE'22*

2022 Sessions Chair, Intl. Conf. on Software Engineering (ICSE)

2022 Sessions Chair, Intl. Conf. on Mining Software Repositories (MSR)

2022 Social Media Chair, Intl. Conf. on Mining Software Repositories (MSR)

2021 Sessions co-host, Intl. Conf. on Automated Software Engineering (ASE)

2020 Social Media Chair, Intl. Conf. on Mining Software Repositories (MSR)

Program Committee

2026 Intl. Conf. on Software Engineering (ICSE), Technical Track

2025 Intl. Conf. on Software Engineering (ICSE), Technical Track

2025 Intl. Conf. on Automated Software Engineering (ASE), Technical Track

2025 ACM Conf. on the Foundations of Software Engineering (FSE), Technical Track

2024 Intl. Conf. on Software Engineering (ICSE), Technical Track

2024 ACM Conf. on the Foundations of Software Engineering (FSE), Technical Track

2024 Intl. Conf. on Mining Software Repositories (MSR), Technical Track

2023 ACM Conf. on the Foundations of Software Engineering (ESEC/FSE), Technical Track
[Distinguished Reviewer Award] 🏆

2023 Intl. Conf. on Software Engineering (ICSE), SEIP Track

2023 Intl. Conf. on Mining Software Repositories (MSR), Technical Track

2023 Intl. Workshop on Natural Language based Software Engineering (NLBSE), Research Track

2022 Intl. Conf. on Mining Software Repositories (MSR), Technical Track

2022 Intl. Conf. on Software Analysis, Evolution, and Reengineering (SANER), ERA Track

2022 Intl. Conf. on Software Engineering (ICSE), SEET Track

2021 Intl. Conf. on Software Maintenance and Evolution (ICSME), Tool Demo Track

2021 Intl. Conf. on Mining Software Repositories (MSR), Mining Challenge Track

2021 Intl. Conf. on Software Analysis, Evolution, and Reengineering (SANER), ERA Track

Journal Review

2022-present Transactions on Software Engineering (TSE), IEEE

2021-present Empirical Software Engineering (EMSE), Springer

2021-present Transactions on Software Engineering and Methodology (TOSEM), ACM

2022-present Journal of Systems and Software (JSS), Elsevier

2024-present Automated Software Engineering (ASE), Springer

2025-present Information and Software Technology (IST), Elsevier

External Conference and Journal Review

2018 Reviewer, Journal of Software: Practice and Experience

2017 Reviewer, ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)

Research Proposal Reviews

2025, 2026 **Panelist and Reviewer**, National Science Foundation (NSF)

2026 **Reviewer**, The Natural Sciences and Engineering Research Council of Canada (NSERC)

2025, 2024 **Ad hoc Reviewer**, National Science Foundation (NSF)

University Service

University and College Service Activities at Drexel University

AY 2026-2027 **Alternate Senator**, Drexel University Faculty Senate, SCIS Representative

AY 2026-2027 **Member**, Drexel University Faculty Senate Policy Review Committee, SCIS Representative

AY 2025-2026 **Alternate Senator**, Drexel University Faculty Senate, CCI Representative

AY 2025-2026 **Member**, Drexel University Faculty Senate Policy Review Committee, CCI Representative

AY 2024-2025 **Alternate Senator**, Drexel University Faculty Senate, CCI Representative

2023 **Review Committee**, Drexel University Provost's Award for Pedagogical Innovation

Department Service Activities at Drexel University

AY 2026-2027 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

AY 2025-2026 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

AY 2025-2026 **Member**, Dept. of CS Programming/Systems Seminar

AY 2024-2025 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

AY 2024-2025 **Member**, Dept. of CS Programming/Systems Seminar

AY 2023-2024 **Committee Member**, Dept. of CS Tenure Track Faculty Recruiting Committee

AY 2023-2024 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

AY 2023-2024 **Member**, Dept. of CS Programming/Systems Seminar

AY 2022-2023 **Committee Member**, Dept. of CS Tenure Track Faculty Recruiting Committee

AY 2022-2023 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

AY 2021-2022 **Committee Member**, Dept. of CS Tenure Track Faculty Recruiting Committee

AY 2021-2022 **Committee Member**, Dept. of CS Software Engineering (SE) Committee

Leadership and other Professional Development

2026 **Invited Panelist**, UR2PhD Undergraduate Mentoring Workshop

2024 **Attendee**, CRA-WP Virtual Career Workshop

2024 **Attendee**, NSF CAREER Workshop, Washington, DC

2024 **Attendee**, CRA Career Mentoring Workshop for early-career faculty

2023 **Attendee**, Drexel CANOPI Workshop: Developing faculty engagement narratives

2023 **Attendee**, Drexel Library Workshop: ChatGPT as a learning tool in higher education

2023 **Attendee**, Drexel TLC Workshop: Practice and Feedback Make Perfect—or at Least Better!

2023 **Invited Speaker**, ACM-W Alumni Panel, University of Delaware

2023 **Attendee**, NSF CAREER Workshop

2022 **Attendee**, ICSE New Faculty Symposium

2022 **Attendee**, NSF CAREER Workshop

2022 **Faculty Volunteer**, Women in Tech Reception, Drexel University

2021 **Panelist and Recruiting Faculty**, CS[-ish] PhD Recruiting Event

2021 **Contributor**, ACM SIGSOFT Empirical Standards for Repository Mining

2021 **Attendee**, ICSE New Faculty Symposium

2021 **Attendee**, NSF Virtual Grants Conference

2020 **Participant**, Grace Hopper Celebration of Women in Computing, University of Delaware

2016 - 2017 **Founder and Chair**, University of Delaware ACM-W Student Chapter

2015 - 2016 **Vice-Chair**, CISters, Women in Computer Science, University of Delaware

2015 **Mentor**, Grace Hopper Celebration of Women in Computing, University of Delaware

2015 - 2016 **Technical Administrator**, Indian Graduate Student Association, University of Delaware

Professional Memberships and Affiliations

- Lifetime Member, Association for Computing Machinery (ACM) *Member since: 2016*
- Member, Institute of Electrical and Electronics Engineers (IEEE) *Member since: 2023*
- Member, Association for Computing Machinery, Special Interest Group on Software Engineering (ACM-SIGSOFT)
- Member, Association for Computing Machinery, Women (ACM-W)

Industry Experience

QA Engineer *Integral Development Corp.*
2013–2014 *Bangalore, India*
Domain: Foreign Exchange Trading

- Performed Web-based Testing, Functional and Regression Testing of in-house Foreign Exchange Trading tool, and ensured adherence to quality norms throughout the testing process.
- Performed debugging of multiple logs of application workflows for failed scenarios.
- Contributed to building the in-house automation test framework using Groovy.

Software Engineer *Theorem Private Ltd.*
2012–2013 *Bangalore, India*
Domain: Computational Advertising

- Single point of contact for the testing team of BRT (Business Response Team), which handled bug/enhancement requests from production. Analyzed software requirement specifications, prepared test plans, test scenarios (using TestLink), created and executed test cases, logged and tracked defects in JIRA and Bugzilla.

- Performed Functional and Regression Testing for the user interface to validate software component functionalities.
- Performed Scoping and estimated the timeline of the project, considering the efforts and number of resources available for a particular production release.

Project Engineer *Wipro Technologies*
2009-2012 *Bangalore, India*
Domain: Online equity trading

- Developed the automation suite for regression testing and debugging of Perl/Shell scripts to run batch processes.
- Created and executed test cases, identified and logged defects/bugs using Clear Quest.
- Performed Sanity testing for new builds & version controlling using Clear-Case.

Domain: Multi Functional Peripheral firmware testing

- Created test plans and test sessions in QC, performed testing on builds - both in development as well as delivery phase using in-house testing tools HP DSS, HP Web Jet Admin.
- Found critical defects through ad-hoc/ exploratory testing, reviewed and fixed bugs in the source code.
- Performed build qualification with Smoke & Sanity testing.

References

Available on request